

Заг. 110 Куты 0 | 1 | 2 | 3

$$Y = \text{N}^{\circ} \text{ куты} \sim \text{Bin}(4, \frac{1}{2})$$

$X = \text{незада}$

X	1000	100	700	250	500
P	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$$\begin{aligned} EX &= (1000 + 400 + 4200 + 1000 + 500) / 16 \\ &= \frac{7100}{16} = \underline{\underline{443,75}} \end{aligned}$$

2. $X_1 = \text{\# тонета го 1-во 1000AB}$

$$\sim \text{Ge}(\frac{1}{16}) \quad EX_1 = 16$$

$X_2 = \text{Обща незада го 1-во 1000AB.}$

Неопределено: $1000 + 15 \cdot \text{\# незада | не сме в 1000AB}$

X	$X \neq 1000$	100	700	250	500
	$\frac{4}{15}$	$\frac{6}{15}$	$\frac{4}{15}$	$\frac{1}{15}$	

$$\Rightarrow E[X | X \neq 1000] = \frac{6100}{15} = 406\frac{2}{3}$$

$$\Rightarrow 1000 + 6100 = 7100.$$

формула: $EX_2 = \sum_k P(X_1=k) E[X_2 | X_1=k]$
 из условия

$$= \sum_k P(X_1=k) E\left[\sum_{i=1}^k \text{результат хода } i \mid X_1=k\right]$$

$$= \sum_k P(X_1=k) \left[\sum_{i=1}^{k-1} E[\text{результат хода } i \mid \text{результат } i \neq 1000] + 1000 \right]$$

$$= \sum_{k=1}^{\infty} \left(\frac{15}{16}\right)^{k-1} \left(\frac{1}{16}\right) \left[(k-1) \cdot \frac{6100}{15} + 1000 \right]$$

$$\sum_{k=1}^{\infty} \left(\frac{15}{16}\right)^{k-1} (k-1) = \sum_{k=0}^{\infty} k \cdot \left(\frac{15}{16}\right)^k = \frac{15/16}{\left(\frac{1}{16}\right)^2} = 15 \cdot 16$$

$$\sum_{k=1}^{\infty} \left(\frac{15}{16}\right)^{k-1} \cdot \frac{1}{16} = \frac{1}{16} \cdot \frac{1}{1 - \frac{15}{16}} = 1$$

$$= \frac{1}{16} \cdot 15 \cdot 16 \cdot \frac{6100}{15} + 1000 \cdot 1$$

$$= \underline{\underline{7100.}}$$

3. Нeka Z_i = сума в клетка i
 $\sim \text{Unif}(\{1000, 1000, 250, 100\})$
 , т.е. $E Z_i = \frac{2550}{5} = 510$.

$$\begin{aligned} \text{Тогава } E \text{rezultat} &= E \sum Z_i \cdot \binom{4}{i} \cdot \left(\frac{1}{2}\right)^i \left(\frac{1}{2}\right)^{4-i} \\ &= E Z_1 = 510, \text{ т.е. } E Z_i = 510 \\ &\quad \text{и } \sum \binom{4}{i} \left(\frac{1}{2}\right)^4 = 1. \end{aligned}$$

Rezultate $\rightarrow 1000 \ 700 \ 1000 \ 500 \ 250$.

$$\begin{aligned} 4. P(\text{га накрая 3 нота в 100}) \\ &= (4p \cdot (1-p)^3)^3 \\ &= 64p^3(1-p)^9 =: f(p) \end{aligned}$$

$$f'(p) = 192p^2(1-p)^9 - 576p^3(1-p)^8$$

$$f'(p) = 0 \Leftrightarrow 192(1-p) = 576p$$

$$\Leftrightarrow 192 = 768p$$

$$\Leftrightarrow \boxed{p = \frac{1}{4}}$$

$$5) \text{ бери: } \frac{\left(\frac{3}{8}\right)^2}{\left(\frac{3}{8}\right)^2 + \left(\frac{8}{27}\right)^2} \approx 62\%$$

Заг. 2 Двеа $Y_1 = X_1 + X_2$ и $Y_2 = X_1 X_2$
с параметри, изотопару с $1/2$.

1. За $X_i \sim \text{Ber}(p)$.

$$X_1 + X_2 \in \{0, 1, 2\} \Rightarrow Y_1 \sim \text{Ber}(p_1)$$

$$P(X_1 X_2 = 1) = p \cdot p = 1 - P(X_1 X_2 = 0)$$

$$\Rightarrow Y_2 \sim \text{Ber}(p^2)$$

2. За $X_i \sim \text{Bin}(n, p)$

$X_1 + X_2$ = сума на n iid $\text{Ber}(p)$
+ сума на n iid $\text{Ber}(p)$
 $\sim \text{Bin}(2n, p)$

Also $X_1 X_2 \sim \text{Bin}(n_2, p_2)$, so T.k.
 $X_1 X_2 \in \{0, 1, \dots, n_1^2\}$ i.e. $n_2 = n^2$.
~~(det p p)~~

$$E[X_1 X_2] = n \cdot p \cdot n \cdot p \stackrel{\text{det p p}}{=} n_2 p_2$$

$$\Rightarrow p_2 = \frac{(np)^2}{n^2} = p^2$$

$$P(X_1 X_2 = n_2) = P(X_1 = n)^2 = p^{2n}$$

$$\stackrel{\text{det p p}}{=} p^{n_2} = (p^2)^{n_2} = p^{2n^2}$$

$$\Rightarrow p^{2n} = p^{2n^2} \Rightarrow p^n = 1 \quad \wedge \quad \exists a \, p_1 \neq 1, a$$

also $X_1 \sim \text{Bin}(n_1, 1)$, so $X_1 X_2 \sim \text{Bin}(n_1^2, 1)$.

3. $\exists a \, X_1 \sim \text{Ge}(p)$. Also $X_1 + X_2 \sim \text{Ge}(p_2)$

T.k. $E[X_1 + X_2] = \frac{2}{p} = \frac{1}{p_2} \Rightarrow p_2 = \frac{p}{2}$

$$D[X_1 + X_2] = \frac{2(1-p)}{p^2} = \frac{1-p_2}{p_2^2} \leq \frac{4-2p}{p^2}$$

$$\Rightarrow 2-2p = 4-2p \quad \swarrow \searrow$$

Also $X_1 X_2 \sim \text{Ge}(p_3)$, i.e.

$$E[X_1 X_2] = \frac{1}{p^2} \Rightarrow p_3 = p^2$$

$$\begin{aligned} \text{Cov}[X_1, X_2] &= E[X_1^2 X_2^2] - (E[X_1 X_2])^2 \\ &= \left(\underbrace{\frac{1-p}{p^2}}_{\text{Var } X_1} + \underbrace{\frac{1}{p^2}}_{(E X_1)^2} \right)^2 - \left(\frac{1}{p^2} \right)^2 \end{aligned}$$

$$= \frac{3-4p+p^2}{p^4} = \frac{1-p_3}{p_3^2}$$

$$= \frac{1-p^2}{p^4}$$

$$\Rightarrow 1-p^2 = 3-4p+p^2 \Rightarrow 2p^2-4p+2=0$$

$\Rightarrow p=1$. Also $X_1 \sim \text{Ge}(1) \Rightarrow X_1 X_2 \sim \text{Ge}(1)$,
where $X_1 X_2 \not\sim \text{Ge}(p_3)$.

4. Also $X_1 \sim \text{Po}(\lambda)$.

$X_1 + X_2 \sim P_0(2\lambda)$ (канпуер
2-рээс тодоромгууц. Бүрх параметер)

Дагумуулам, $X_1 X_2 \sim P_0(\lambda^2)$

$$E X_1 X_2 = \lambda^2 \Rightarrow \lambda^2 = \lambda^2$$

$$D X_1 X_2 = E X_1^2 X_2^2 - (E X_1 X_2)^2$$

$$= (\lambda + \lambda^2)^2 - \lambda^4$$

$$= \lambda^2 + 2\lambda^3 = \lambda^2 = \lambda^2$$

$$\Rightarrow 2\lambda^3 = 0 \quad \swarrow \searrow$$

зөг. 3 1. $E f = 3\mu$
 $D f = 3\sigma^2$

Хэрэв $f_{\sigma} = \frac{1}{4} (f(X_1, X_2, X_3) + f(X_2, X_1, X_3) + f(X_3, X_2, X_1) + f(X_1, X_3, X_2))$

$$f_{\sigma} = \frac{1}{4} (4(X_1 + X_2 + X_3)) = f$$

$$\Rightarrow E f_{\sigma} = E f \quad \text{u} \quad D f_{\sigma} = D f$$

$$2. f = \frac{1}{4} (X_1 + X_2 - X_3 + X_3 + X_2 - X_1 + X_2 + X_1 - X_3 + X_1 + X_3 - X_2)$$

$$= \frac{1}{4} (2X_1 + 2X_2) \Rightarrow$$

$$\Rightarrow \mathbb{E}f = \mathbb{E}f = \mu; Df = D(X_1 + X_2 - X_3) = 3\sigma^2$$

$$Df = D\left(\frac{X_1 + X_2}{2}\right) = \frac{1}{4} (DX_1 + DX_2)$$

$$= \frac{\sigma^2}{2} \Rightarrow \text{у нас эффект.}$$

высказана в с помощью
гипотезы.

3. Если

$$Y_1 = X_1 \cdot X_2 - X_3$$

$$Y_2 = X_3 \cdot X_2 - X_1$$

$$Y_3 = X_2 \cdot X_3 - X_1 = Y_2$$

$$Y_4 = X_1 \cdot X_3 - X_2$$

$$\mathbb{E}f = \mathbb{E}Y_1 = \mu^2 - \mu = \mathbb{E}Y_i$$

$$Df = D(X_1 X_2) + DX_3 = \mathbb{E}X_1^2 X_2^2 - (\mathbb{E}X_1 X_2)^2 + DX_3$$

$$= (\sigma^2 + \mu^2)^2 - (\mu^2)^2 + \sigma^2$$

$$= \sigma^2 (\sigma^2 + 2\mu^2 + 1).$$

$$E f_{\sigma} = \mu$$

$$D f_{\sigma} = D \left(\frac{1}{4} (Y_1 + Y_2 + Y_3 + Y_4) \right)$$

$$= \frac{1}{16} D (Y_1 + 2Y_2 + Y_4)$$

$$= \frac{1}{16} (D Y_1 + D Y_2 + D Y_4 + 10 \operatorname{Cov}(Y_1, Y_2))$$

$$= \frac{3}{8} D Y_1 + \frac{5}{8} \operatorname{Cov}(Y_1, Y_2)$$

$$\Rightarrow D f - D f_{\sigma} = \frac{5}{8} (D Y_1 - \operatorname{Cov}(Y_1, Y_2))$$

и ако показател $\varepsilon > 0$, то

f_{σ} е по-хороша оценка за μ :

$$\operatorname{Cov}(X_1 X_2 - X_3, X_3 X_2 - X_1)$$

$$= \operatorname{Cov}(X_1 X_2, X_3 X_2) - \operatorname{Cov}(X_3, X_3 X_2)$$

$$- \operatorname{Cov}(X_1 X_2, X_1) + \operatorname{Cov}(X_3, X_1)$$

$$= \operatorname{Cov}(X_1 X_2, X_2 X_3) - 2 \operatorname{Cov}(X_1, X_1 X_2)$$

$$= E X_1^2 X_2 X_3 - (E X_1 X_2)^2 - 2(E X_1^2 X_2 - E X_1 E X_1 X_2)$$

$$= \mu^2(\sigma^2 + \mu^2) - \mu^4 - 2((\sigma^2 + \mu^2)\mu - \mu^3)$$

$$= \mu^2\sigma^2 - 2\mu\sigma^2 = \sigma^2(\mu^2 - 2\mu)$$

$$\Rightarrow Df - Df_\sigma = \frac{5}{8}\sigma^2(\sigma^2 + 2\mu^2 + 1 - \mu^2 + 2\mu)$$

$$= \frac{5}{8}\sigma^2(\sigma^2 + \mu^2 + 2\mu + 1)$$

$$> 0 \quad .$$